Financial Management (MBAZG521)

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Financial Statement Analysis

1.2 Key Financial Statements

Managers, shareholders, lenders, suppliers, customers, tax authorities, and other interested groups seek answers to the following important questions about a firm:

- What is the financial position of the firm at a given point of time?
- How has the firm performed financially over a given period of time?
- What have been the sources and uses of cash over a given period of time?

Key Financial Statements

To answer the above questions, Financial Accounting prepared the following three key statements

- **Balance Sheet:** Shows the financial position of the firm at a given point of time
- **Profit and Loss A/C:** Reflects the performance of the firm over a period of time
- **Cash Flow Statement:** Displays the sources and uses of cash during the period.

Financial Statements

- Provide information on how the firm has performed in the past and what is its current financial position.
- Serve as a convenient device for the stakeholders to set performance norms and impose restrictions on the management of the firm.
- Provide convenient templates for financial forecasting and planning.

Balance Sheet

Total Assets = Total Liabilities + Owner's Equity

TA = TL + OE

Total Assets = Total Fixed Assets + Total Current Assets

Total Liabilities = Total Long-term Liabilities + Total Current Liabilities

Owner's Equity = Shareholders' Funds + Reserves and Surplus

Balance Sheet Format

Liabilities	Amt	Amt	Assets	Amt	Amt
Capital Share capital Retained earnings			***Fixed Assets (less Depreciation)*** Land and Buildings Plant and Machinery Equipments		
Long-term Liabilities Debentures Bonds Long-term loans Note			***Current Assets*** Cash Bank Marketable securities Accounts receivables (Debtors) Inventories Prepaid expenses		
Current Liabilities Accounts payables (creditors) Taxes payables Accrued expenses Deferred revenues Current portion of long-term debt			***Other Assets*** Investments Intangible assets		

Profit and Loss A/C

Profit After Tax (PAT) = Revenue – Expenses

Based on the two important principles of Accounting

- **Realization Principle:** Revenue is recognized when the transaction generating the revenue takes place
- **Matching Principle:** The expenses associated with a product/service are recognized when the product/service is sold

Profit and Loss A/C Format

Particulars	Amount	Amount
Sales		XXX
(less) Cost of Goods Sold		XXX
Gross Profit		XXX
(less) R & D Expenses	XXX	
Sales & Distribution Expenses	XXX	
Administration and General Expenses	XXX	XXX
Operating Income (EBDIT)		XXX
(less) Depreciation		XXX
EBIT		XXX
(less) Interest		XXX
EBT		XXX
(less) Tax		XXX
EAT or PAT		XXX

Cash Flow Statement

The Cash Flow Statement reflects an enterprise's major sources of

- Cash Receipts, and
- Cash Payments.

It reports the cash effects during a period of an enterprise's

- Operations,
- Investing transactions, and
- Financing transactions

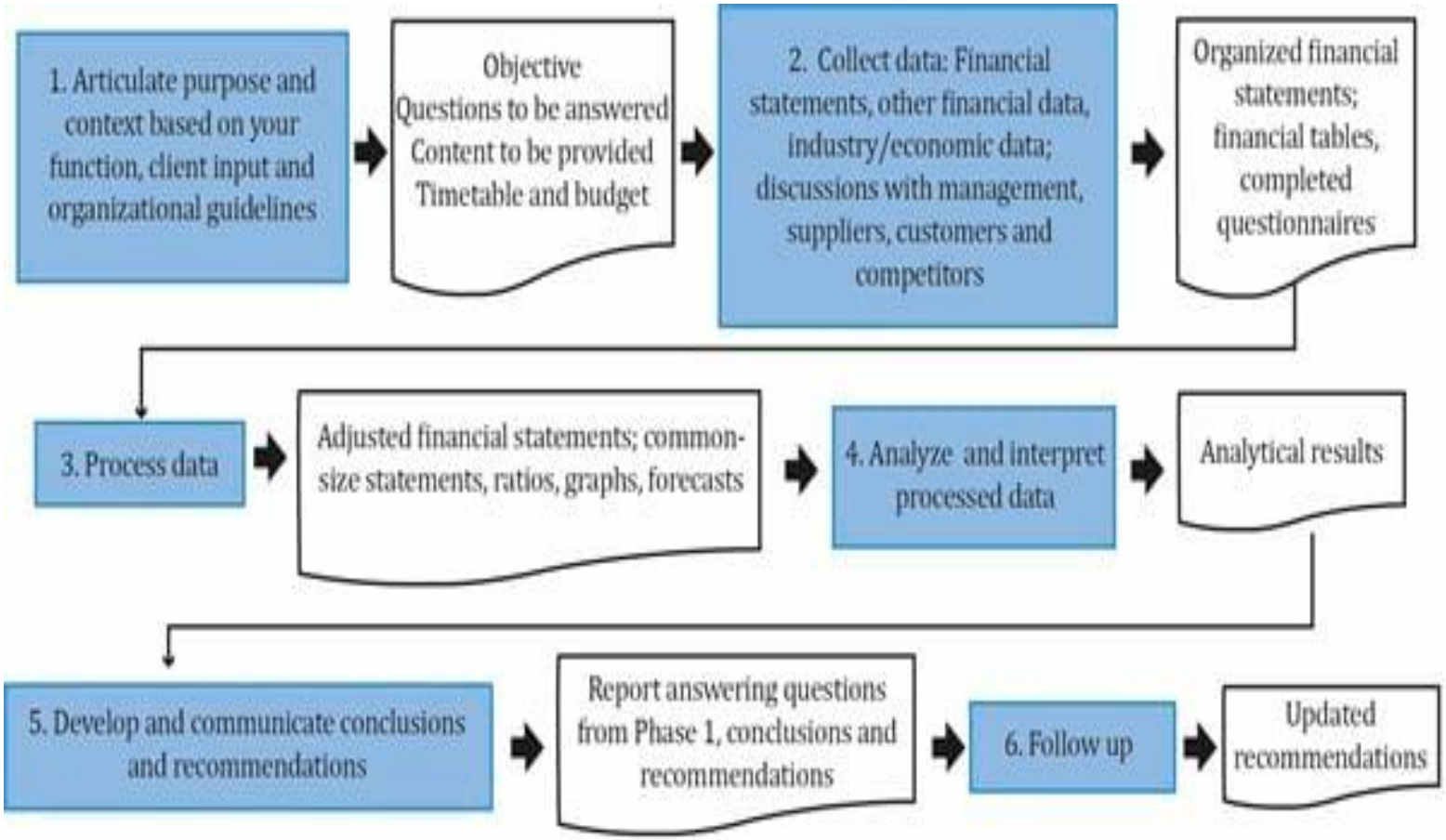
It focuses on

- Aggregate of cash and
- Cash equivalents

Cash Flow Statement Format

SN	Particulars	Amount	Amount
(a)	Cash Flow from Operating Activities List of individual inflows and outflows Net cash provided by/used in Operating Activities	XXXX	XXXX
(b)	Cash Flow from Investing Activities List of individual inflows and outflows Net cash provided by/used in Investing Activities	XXXX	XXXX
(c)	Cash Flow from Financing Activities List of individual inflows and outflows Net cash provided by/used in Financing Activities	XXXX	XXXX
(d)	**Net increase/decrease in cash (a+b+c)**		**XXXX**
(e) (f)	**Cash and cash equivalents, beginning of period** **Cash and cash equivalents, end of period (d+e)**		**XXXX** **XXXX**
	Schedule of non-cash investing and financing activities List of individual transactions		

Financial Statement Analysis Framework



Source: www.ift.world

1.3 Financial Statement Analysis Techniques

- **Financial Ratio Analysis (including Dupont analysis)**
- **Time Series of Financial Ratio (Trend Analysis)**
- **Percentage Format Financial Statement (Common-size financial analysis)**
- **Industry benchmarking Analysis**
- **Cost-volume-profit analysis**

Financial Ratio Analysis

1. Liquidity Ratio
2. Solvency or Leverage Ratio
3. Turn over Ratio or Efficiency Ratio
4. Profitability Ratio
5. Valuation Ratio

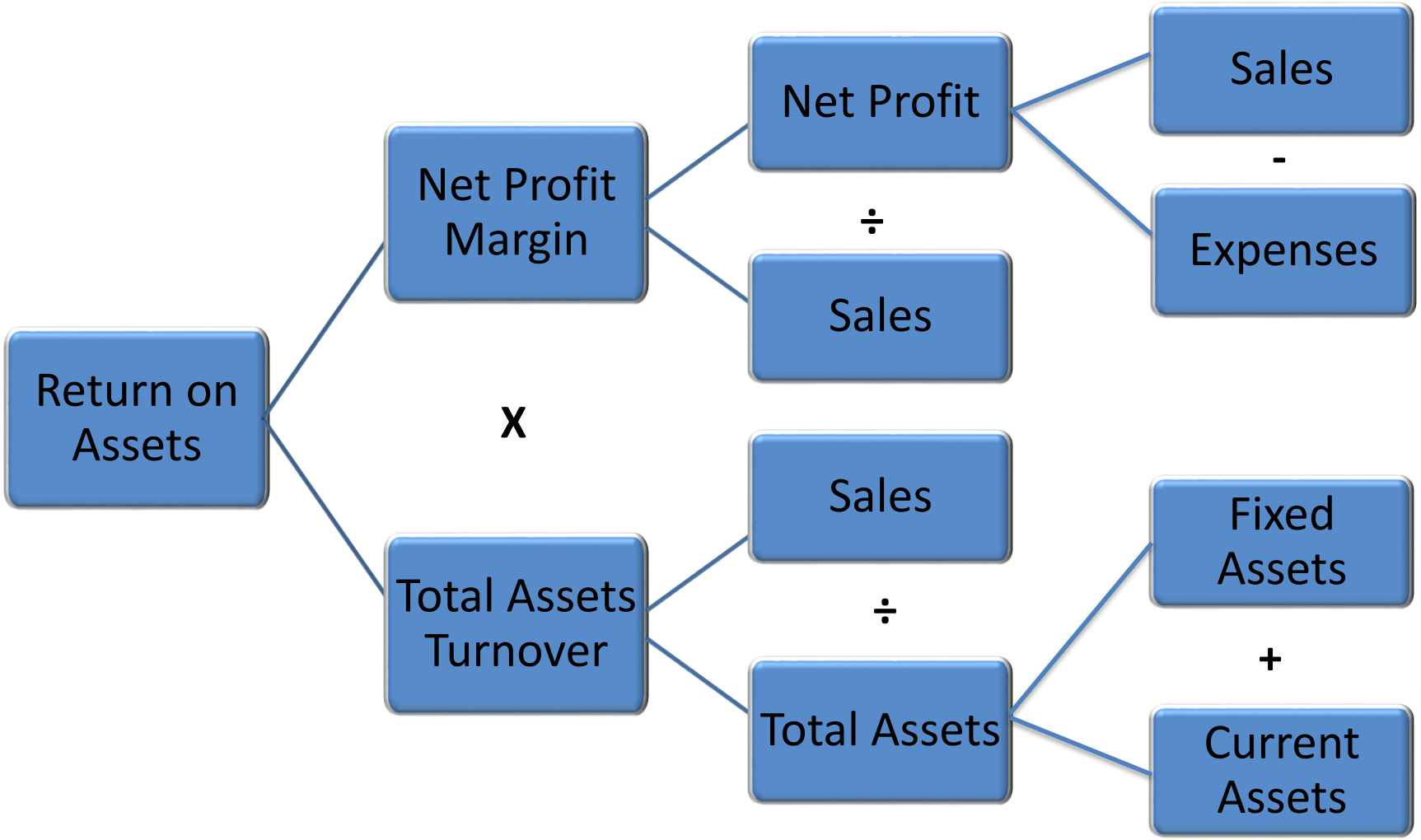
Financial Ratio Analysis

Ratio	Formula
Liquidity Ratio	
• Current Ratio	Current Assets/ Current Liabilities
• Acid-test Ratio	Quick Assets/Current Liabilities
Leverage Ratio	
• Debt-equity Ratio	Total Liabilities/Shareholders Funds
• Interest Coverage Ratio	EBIT/Interest
Turn Over Ratio	
• Inventory Turnover Ratio	Sales/Inventory OR COGS/Inventory
• Debtors' Turnover Ratio	Net Credit Sales/Trade Receivables
• Fixed Assets Turnover Ratio	Sales/Net Fixed Assets
• Total Assets Turnover Ratio	Sales/Total Assets

Financial Ratio Analysis

Ratio	Formula
Profitability Ratio	
• Gross Profit Margin Ratio	Gross Profit/Sales
• Net Profit Margin Ratio	Net Profit/Sales
• Return on Assets (ROA)	PAT/Total Assets OR [PAT + Interest (1-Tax rate)]/Total Assets
• Earning Power	EBIT/Total Assets
• Return on Capital Employed	EBIT (1-T)/Total Assets
• Return on Equity (ROE)	Equity Earnings/Equity or PAT/Equity
Valuation Ratio	
• Price-Earning Ratio (PE)	Market Price per Share/Earning per Share
• Yield	(Dividend + Price Change)/Initial Price
• Market Value to Book Value	Market price per share ratio/book value per share

DU PONT Analysis



Time Series of Financial Ratio

- Besides looking at the ratios for one year, look at the ratios for several years
- Helps to detect secular changes and avoid the bias introduced by transitory forces.

Percentage Format Financial Statements

- To understand better a firm's changing financial position and performance over time
- To compare one firm with another

In two forms

- Horizontal : Percentage of previous year
- Vertical also known as Common Size Statement Analysis
  - Balance Sheet: Each asset and liability amount is expressed as percentage of total assets
  - Profit and Loss A/C: Each item is expresses as a percentage of total revenues.

Industry Benchmarking Analysis

- Technique to use to compare your business to a competitor business or to businesses in the industry at large.
- It can help answer questions like:
  - Does your business have a competitive advantage?
  - Are there weaknesses or inefficiencies that we can address and close the gap between your company and competitors?”

Cost-volume-profit Analysis

- Set relationship between sales, production volume and business profit
- Helps to set break-even point, target sales, production and profit
- Analyse change in fixed cost, variable cost with the change in target sales

Time Series Analysis

Module - 5

Limitations in Financial Statement Analysis

- Heuristic and intuitive characters
- Development of benchmarks
- Window dressing
- Price level changes
- Variation in accounting polices
- Interpretation of results

Widely used software for FSA

- Microsoft Excel – VBA Macro tool
- MATLAB
- SAS
- Quilkview
- ORACLE
- Python, R

Ratio Analysis: Practice Exercise

ABC Ltd. has made plan for the next year. It is estimated that the company will employ total assets of Rs. 800000, 50% of the assets being financed by borrowed capital at an interest rate of 16% per year. The direct costs for the year are estimated at Rs. 480000 and all other operating expenses are estimated at Rs. 80000. The goods will be sold to customers at 150% of the direct costs. Income-tax rate is assumed to be 50%. You are required to calculate

1. Net profit margin
2. Return on assets
3. Assets turnover and
4. Return on owner's equity

THANK YOU!!!